A CONSTRUCTED FAR END

BY

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THE METHOD APPLIED TO IDEAS
DELTA NULL, WHAT A BOUNDARY HOLDS, AND HYPOTHESES WITH THEIR TESTS

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

Every measurement in this body of work reads an object against a reference built from the object's own parts: the most disordered arrangement its counts allow. This research paper sets out that method and applies it to ideas instead of corpora. A reference of this kind is constructed and laid over a real system, and it has to be allowed instead of derived, as the oldest working constructions in mathematics were. Two practices follow. An instrument is run on inputs built to make it fail, since one that cannot be made to say no reports nothing when it says yes. And a possibility that reads as incoherent is allowed and pushed until it stops cohering, since what reads as sense to a person and what a structure permits are measured here to come apart.

The delta null idea is the worked case. Of its three claims, invariance under translation, permutation and change of coordinate representation holds over 3,421 bench rows and 465,546 true occurrences, and permanent exclusion on hitting the boundary holds by construction. Recovering a volume from its boundary fails, because the converse of exclusion fails for every proper subset of a pattern's points: the boundary refuses and does not predict. A closed boundary holds a number of separable readings set by its measure over the finest resolution, independent of shape and topology, which mode counts on spheres and integer-point counts on flat tori confirm against Weyl's law to within 0.0122 percent; what the interior holds is a separate question that nothing here tests. Four entries of a notebook of suppositions, and one proposal beside them, are stated as hypotheses with the test that would decide each. The notebook itself, and the posits with no test, are kept whole as wants in the workbook.

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1

A CONSTRUCTED FAR END

The measure used throughout this work places an object at one end and the most disordered arrangement of its own parts at the other, and reads the distance between them. The far end is the maximum entropy arrangement consistent with the object's counts, the reference that Jaynes made the basis of statistical inference [8]. The reading is that distance and never a value on its own. A dispersion of 0.28 says nothing without the reference it was taken against.

1.1 A REFERENCE LAID OVER A SYSTEM

The far end is constructed and then laid over the real system, and that overlay is an old working method in mathematics. A square root of minus one, laid over algebra, gives alternating current and quantum mechanics. Parallel lines that meet, laid over geometry, lead to the geometry of general relativity. A quantity both zero and not zero gives calculus. A function zero everywhere whose integral is one gives signal processing. Subtracting one infinity from another, which Feynman called hocus-pocus [17], gives quantum electrodynamics. Each met objection when it was introduced, and each was kept for what the overlay predicted.

Every null in this work is that move. A permutation null is a corpus nobody wrote, laid over one somebody did, and the difference is the reading. Scattering a protein's atoms through its own bounding box builds a protein that cannot exist. The GF(2)-linear variant of SHA-256 is that function with the part that makes it that function removed, and overlaying it gives an anchor that algebra pins at 0.50000000 [21]. A far end cannot be derived from the object, and it has to be allowed instead of argued for.

1.2 BUILDING THE THING EXPECTED TO FAIL

A null that has never fired is not a null. The memoryless arms are run on every sweep and return 0.99 to 1.01 every time. A flat corpus of one byte repeated is put through five instruments at once. A generated ramp built with a width of 512 is measured and returns 511. The counterexample class that took back three of the four regularities recorded in the workbook [20] was built the same way. An instrument that cannot be made to say no is not reporting anything when it says yes.

1.3 COHERENCE TO A READER IS NOT COHERENCE IN THE INFORMATION

What reads as sense to a person and what a structure permits come apart, and the workbook puts numbers on the gap [20]. A perfect lattice carrying random values departs from neither null, because a survivor needs a displacement and a value together and random values supply no repeated pairs. A counter of period sixteen has a collision entropy of $\log_2 16 = 4.0$ bits. The quantity that governs cost is invariant under permutation and calls a perfectly regular corpus memoryless, since it cannot see an arrangement at all. English prose measures 3.764 bits against a memoryless control at 3.779. A memoryless process carrying a delimiter reaches a brevity correlation of -0.401, stronger than any natural text measured.

Each of those is incoherent to a reader and coherent as information, and intuition has no traction on any of them. A possibility that reads as absurd is therefore allowed instead of dismissed, and allowed so it can be pushed until it stops cohering. The place it stops is the finding. Nothing in this research paper is believed on the strength of its plausibility, and nothing is dismissed for the lack of it. The next chapter is the worked case.

2

DELTA NULL AS A BOUNDARY

2.1 THE MAPPING

The ground set Σ has no topology, metric or order. It supplies only distinctness: two symbols are equal or they are not. The measurement δ_0 is the indicator of equality, and it turns a comparison at a displacement $d \in G$ into one bit, match or no match. The anchor set $A \subset D$, a subset of a pattern's points, acts as a boundary between the candidate alignments and the rejected ones.

Checking $\mathrm{Anc}_A(s)$ decides whether an alignment $s \in G$ lies inside the feasible region or crosses the boundary into the rejected exterior. The contrapositive $\neg\mathrm{Anc}_A(s) \Rightarrow \neg\mathrm{Occ}(s)$ states that hitting the boundary once excludes s permanently. The choice of A is free with respect to correctness, and the boundary governs only cost: the density and entropy of Σ over A set how many candidates the filter rejects per operation. Because the boundary is defined through displacements in G and equality in Σ alone, the search should be invariant under translation, permutation and change of coordinate representation.

2.2 THREE CLAIMS, AND WHERE EACH STANDS

Invariance under translation, permutation and change of coordinate representation. Measured, and it holds. The lattice bench sweeps exactly this: 3,421 rows over alphabets of 2 to 256 symbols, patterns of 1 to 32 points, dimensions 1 through 8, a rotated point set, a scatter no rectangle covers, a complex alphabet with irrational parts compared over its storage, and an order check against a permuted base list. Of 465,546 true occurrences, none is refused. The core takes a base list, a displacement list and a callback answering whether two positions carry the same symbol. It has no dimension parameter and no symbol type, and a core that cannot see either cannot depend on either.

Hitting the boundary once excludes the point permanently. Proved, and checked on every row. This is the first proposition of the permutation null [19]: any subset of a pattern's points is a necessary condition for an occurrence. A failure at one anchor refutes the alignment and no later evidence restores it. Every bench that runs an arm against the exact compare checks it.

The volume is recoverable from its boundary. Fails. The second proposition answers it: the converse fails for every proper subset of the points [19]. The exact compare is irreducible, and every arm has to verify what its anchors admitted. A boundary that excludes is not a boundary that determines.

2.3 THE DEMON REFUSES AND DOES NOT PREDICT

The strong form of the idea, a Laplace's demon that recovers the interior from the boundary, fails. The weak form survives, and it is the algorithm: an anchor set is a boundary, crossing it is final, and nothing about crossing it says what lies on the other side. A filter that refutes cheaply and determines nothing is the shape the whole construction has. It is the reason soundness costs nothing while completeness costs a full compare.

2.4 WHAT IS NOT TESTABLE HERE

The holographic principle in physics concerns a bound on the information a region can hold [1, 11], in line with the entropy bound [2]. Nothing in this repository measures that, and the resemblance to an anchor boundary is an analogy. The analogy earns its place only in the three claims it produced, two of which hold and one of which fails.

3

WHAT A BOUNDARY HOLDS

A closed surface holds a number of separable readings set by its own measure divided by the finest detail reaching it, raised to the dimension of the surface. Shape does not enter. Topology does not enter. Only the measure and the resolution do.

Two consequences follow. What a region holds scales with the area of its boundary and never with its volume. And a source at depth d inside a ball of radius R, at radius $r = R - d$, reaches harmonic degree l as $(r/R)^l$. Depth sets bandwidth, and a source prints a patch of angular size about d/R however small the source itself is. A spot on a boundary is always wider than the thing that made it.

3.1 DEPTH SETS BANDWIDTH

Under diffusion the boundary response to one interior source at radius r is the Poisson kernel, and its Legendre expansion carries $(2l + 1)(r/R)^l$ at degree l. Everything else follows from that line. A source at the middle reaches degree zero alone and spreads over the whole sphere. A source against the shell reaches every degree and prints a small bright spot.

Reading the depth back out of a spectrum is a slope: dividing out the shape term and the conduction term leaves a straight line in degree whose gradient is twice the logarithm of r/R. Measured against sources placed at known depths, the recovery returns 0.3000, 0.5500, 0.8000 and 0.9200 for radii of 0.30, 0.55, 0.80 and 0.92. It holds under conduction as well, and that check detects an exponent applied twice.

3.2 THE AREA LAW, COUNTED EXACTLY

Degrees up to l carry $(l+1)^2$ modes on an ordinary sphere, and the general count in n dimensions is a sum of two binomial coefficients. That count is a whole number with a formula. It needs no sampling and no rounding.

Weyl's law [14] predicts the same count from the surface measure alone: the volume of the unit ball in the boundary's own dimension, times the boundary's measure, times the eigenvalue to half that dimension, over two pi to that dimension. The two sides share no term. Compared at 2, 3, 5, 8, 12, 19, 24 and 40 dimensions, they agree to within 0.0122 percent, and at forty dimensions the exact count is a 157 digit integer.

The residual is a remainder, never a disagreement. At nineteen dimensions it runs 0.1131, 0.0282, 0.0070 and 0.0018 percent as the degree quadruples each time. An error term of order one over the degree falls by four for every quadrupling.

3.3 THE FLAT TORUS, AND WHY IT MATTERS

Spherical harmonics settle the dimension half of the claim, alone, because they live on spheres. A cube's surface has no such formula, which leaves shape resting on packing.

A flat torus has one. Glue opposite edges of a square and the eigenvalues of the Laplacian become $(2\pi/L)^2$ times the squared length of an integer vector. Counting modes below a cutoff is counting integer points inside a ball. The count is an integer, it holds no transcendental, and it is available at any dimension by raising the every-square series to a power and adding up its coefficients.

This is worth having because a flat torus is not a sphere in any respect that could smuggle the answer in. It is flat where the sphere is curved. It has a hole where the sphere has none. Its symmetry group is a lattice where the sphere's is a rotation group. The same constant comes off both, exactly, with nothing sampled.

It also puts the remainder into known company. The difference between the exact lattice count and the volume of the ball is the lattice point problem, the Gauss circle problem in two dimensions. For balls in dimension $d \geq 5$ the remainder is settled at order R^{d-2} [13], dimension four is settled up to a logarithm, and dimensions two and three are open. The leading term of the area law is settled here and the remainder belongs to the lattice point problem.

3.4 WHY SHAPE CANNOT ENTER

A seam where two edges meet contributes nothing to a surface measure. Neither do a cube's edges, its corners, or the point where three facets meet. Every one of them is measure zero on the surface. None of them can move a count that depends on measure alone.

Shape does not enter for that reason, and it is not a coincidence to be confirmed one shape at a time. A cube differs from a sphere precisely in its edges and corners, and those are exactly the parts that carry no measure. The count cannot see them. Overlap would be the exception, since overlap has positive measure and would count the same area twice.

The same rule governs the packing that measures this. Packed points may touch and may not overlap, and tangency is the densest limit. Touch freely, never overlap, and the measure is untouched either way.

3.5 DIFFUSION IS DWELL

Diffusion is not a blur applied to a picture. It is how long the light stayed inside before it left, and both readings come off one medium sorted by arrival.

Light that leaves early went nearly straight, still knows what it passed, and prints an edge. Light that leaves late has turned over so many times that it arrives from every direction at once. It

fills the shadow it would otherwise have cast and carries how much in place of where. The same photons either way. A clock decides which of them are being counted.

That split is worth more than either half. A steady reading cannot separate absorption from scattering: a weakly absorbing dense medium and a strongly absorbing sparse one give the same number. Adding the clock breaks the degeneracy. The earliest arrivals give the shortest path and the straight-line geometry, the mean arrival gives the absorption, the spread of arrivals gives the scattering, and the tail gives the deepest excursions.

3.6 THE SMEAR

From inside, dwell is a tail behind a moving source. From outside there is no tail to see, because the inside is not what an outside observer has. What they have is the surface, and the tail arrives there as a smear.

The smear is worth more than the spot. A spot says where something is. A smear says where it was going and how long it took: its length is the dwell and its direction is the direction of travel, and both come off a surface that never moved.

3.7 THE CONDITIONS

The account needs four things and fails without any of them.

Closed. An open surface bounds no region. There is no interior to determine and flux leaves unaccounted. A surface pinched at isolated points and open between them fails on both counts: it is not a manifold at the pinch and it is still bounded everywhere else.

Orientable, for containment only. Weyl's law holds on any compact surface, orientable or not. The counting survives. What needs orientability is the outward normal, without which inside and outside are not defined. A non-orientable boundary keeps its capacity and loses its containment.

A gradient. No gradient, no flux. No flux, and the boundary records a uniform nothing however long it is watched.

Coherence. A hologram works by interfering a coherent field with a reference, which records phase as intensity on a two dimensional plate. Without coherence only intensity survives, and recovering the interior from intensity alone is the phase retrieval problem [3, 7]: non-unique in one dimension, generically unique and badly conditioned above it. Coherence is the difference between an inversion that is linear and a guess that is not.

Coherence also supplies the length the count was missing. The number of readings is unbounded as the resolution improves and finite at any resolution. The bound only becomes a number once something fixes a shortest length. In an optical setting the coherence length is that length, and the capacity is the area over the coherence area.

3.8 WHAT IS NOT ESTABLISHED

Anisotropy. The boundary map on an anisotropic medium is invariant under any diffeomorphism fixing the boundary [9], and a family of interiors then gives identical boundary data. Motion alone does not rescue this, since transforming the medium and its contents together leaves the boundary untouched. Motion under a law does: a diffeomorphism relabels space, straight lines curve, Keplerian orbits stop obeying Kepler, and only the untransformed world satisfies the law it was supposed to obey. Recovering orbital radii from a boundary light curve rests on exactly this, and the work is being done by Kepler and not by geometry.

Stability. Uniqueness is not stability [10]. Recovering detail at scale x needs precision growing exponentially in $1/x$. Everything is recoverable in principle and buried under noise in practice. The spectrum of a real measurement dies by degree 46, and past there the inversion divides by a gain already below one part in a million.

Shape above five dimensions. Verified at three, four and five dimensions, where sphere, cube and cross-polytope give the same constant to within a spread of 1.03, 1.07 and 1.10. Above that the count has to be packed, and packing stays faithful only while the gap is well under the radius of curvature of the surface. Holding to that limit at nineteen dimensions needs candidates in the hundreds of millions against a kept set in the hundreds of thousands.

The constant out front. The scaling is provable and shape independent. The constant depends on packing density, which is known to be optimal in dimensions one, two, three [6], eight [12] and twenty-four [4] and open everywhere else, dimension nineteen included. The measurements here use a greedy packing, applied identically on every shape. It measures shape dependence and never packing efficiency.

3.9 THREE ERRORS THAT PRODUCE PLAUSIBLE NUMBERS

Each of these produces a plausible number and none announces itself.

A half-integer arm of the gamma function written with $(2j+1)/2$ in place of $(2j-1)/2$ is wrong at every odd argument. At $n = 2k+1$, $\Gamma(n/2) = \sqrt{\pi}\prod_{j=1}^{k}(2j-1)/2$, and the wrong factor makes it larger by $2k+1$: three at $n=3$, five at $n=5$. The measure of S^{n-1} is $2\pi^{n/2}/\Gamma(n/2)$, and it comes out too small by the same factor at every odd dimension. The constant then alternates high and low with the parity of the dimension. The cube never touches the gamma function and stays flat, which locates the error. A count computed from the area and then checked against the area agrees with itself and finds nothing.

A limit on how wide a packing gap may be, written down and then ignored while picking gaps that saturate quickly, spreads the constants by three, seven and fourteen times at twelve, sixteen and nineteen dimensions. That spread reads exactly like shape dependence. It is one gap being too wide, on every shape at once.

A remainder tested for the wrong behavior fails a correct count. The error in a lattice count oscillates while its envelope falls, and requiring each radius to beat the one before it rejects a correct count. Measured over a band of radii instead of at a point, the envelope comes down by a factor of forty-eight.

3.10 THE DEMON, PRICED

Laplace's demon knows the state of everything and computes the rest. On the standard reading of quantum mechanics, indeterminacy refutes it qualitatively. What the measurements here do is price the classical failure. That is the more useful statement.

A demon reading a boundary gets the measure over the resolution, raised to the dimension of the boundary. That is finite at any resolution and unbounded across resolutions. Nothing stops it improving except the cost of improving. The cost is exponential: recovering detail at scale x needs precision growing like the exponential of $1/x$, because the inversion divides by a gain falling as $(r/R)^l$. The spectrum of a real measurement dies by degree 46, and past there the division is by a number already under one part in a million. The demon meets no wall of principle. It meets a bill.

The distinction that matters cuts against the reading this work invites. Verifying the area law establishes that the channel is that wide. It does not establish that the interior holds only that much. Counted the ordinary way, a volume carries $(R/x)^n$ degrees of freedom while its boundary carries $(R/x)^{n-1}$, leaving the boundary short by a factor of R/x. The holographic principle [1, 11] says the interior really is only boundary-sized and nothing is lost, in line with the entropy bound [2]. That is a physical conjecture and nothing here tests it. The pipe is measured. What has to fit through it is not.

The fork is sharp. If the holographic principle holds, a demon at the boundary has exactly enough and its only difficulty is precision. If it does not, the demon is short by R/x however precise it becomes, and a counting deficit is not a rounding one. No arithmetic buys it back.

A third pressure applies either way. The demon sits inside the region it computes [15]. Its own state is written on the same boundary and counted against the same budget. A demon storing a description of the interior has to store it somewhere the interior can already see.

3.11 READING THE DIMENSION OFF A LINE

A periodic structure in n dimensions, cut at an irrational angle and projected into fewer, comes out quasiperiodic: never one period, but several with irrational ratios between them. Penrose tilings are a five dimensional lattice seen in two [5]. The correspondence runs both ways for a pattern with finitely many rationally independent frequencies, which lifts to a periodic lattice in high enough dimension. It is not a property of aperiodic order in general.

That turns into a measurement. When the cut is totally irrational, the components of its direction independent over the rationals, the count of rationally independent periods in a one dimensional reading equals the dimension of the lattice it was cut from; a cut with a rational relation among them lies in a rational subspace of lower dimension, and the count reports that dimension. Cutting a square lattice at the golden slope, which is totally irrational, and reading it back gives two gap lengths and never three, their ratio the golden mean to one part in ten to the fourteenth, their counts to four parts in ten thousand, no period at any length up to half the sequence, and every strong peak named by a pair of whole numbers against two generators. Against one generator the worst peak misses by 0.382 of it. Against two, by six parts in ten thousand.

A boundary reading can therefore report the dimension of a structure it never had access to, by counting how many of its periods refuse to be multiples of one another.

3.12 A FOURTH ERROR: THE WRONG QUANTITY KEPT SMALL

The packing gap has to be small against the shape's own geometry. Holding it instead to a fraction of the typical distance between two points on the surface is a different requirement in high dimensions.

On S^{n-1} the squared distance $2 - 2\langle x, y\rangle$ between two uniform points has mean 2 and standard deviation $2/\sqrt{n}$, and the distance concentrates near $\sqrt{2}$ as the dimension grows. A gap fixed at half of that is 0.707 of the radius of curvature, and a cap that wide is nothing like the flat disc the count assumes it is. A cube's facets are flat and suffer none of it, and the two shapes part company and the answer reads as shape dependence. That criterion passes a twelve dimensional run whose constants spread by 2.1 times: sphere 5.37, cube 11.41, cross-polytope 8.20.

Measured against the shape instead, at a gap well under every radius in play, the same three shapes give 0.664, 0.644 and 0.650 at three dimensions, 0.660, 0.619 and 0.640 at four, and 0.689, 0.627 and 0.658 at five. Spread 1.03, 1.07 and 1.10.

The shape half of the claim is verified there, and it is also the end of what packing reaches. Five dimensions take ten million candidates for twenty thousand kept points, and the count needed to stay in the flat regime grows the way a packing number does. No device makes that cheap at nineteen dimensions. The limit is structural and never a compute budget, and the exact routes on a sphere and on a flat torus are what reach up there.

3.13 THE NESTING, BOTH DIRECTIONS

Every boundary is inside another one, and everything a boundary holds is a boundary of its own. The outside of a shell is the inside of the next shell out, and the contents of a shell are shells with contents. Running outward: the cloud is inside its shell, the shell is inside the room, the room is inside whatever holds the room, and in this case what holds it is a universe. Running inward: each light is a closed surface, its own constituents are confined by it, and their traces smear across it exactly as the cloud's traces smear across the shell.

Neither direction terminates on its own, and both terminate for the observer.

Downward it stops at resolution. A constituent's smear on its own small surface subtends less than the finest detail the boundary between it and the observer can carry. Below that it is not there to be seen, and drawing it anyway would be inventing detail no boundary holds. The recursion stopping where resolution stops is the area law applied to the picture instead of stated about it.

Upward it stops at whatever the observer is inside. There is no vantage point outside everything, since standing outside one boundary places the observer inside the next. Stepping back from a wall does not escape a boundary; it changes which boundary is being read.

One level is measured. A source at radius r reaches degree l as $(r/R)^l$, the count on that boundary follows the area law, and the same constant comes off different shapes. That the arrangement repeats at every scale is a structural claim about the shape of the account and never a result taken from it. It follows if the same physics holds at every scale, and that premise is assumed and not measured.

4

HYPOTHESES FROM THE NOTEBOOK

A notebook of suppositions was written beside this work, in the order the thinking happened. It is kept whole, with the posits that have no test, as wants in the workbook [20]. Four of its entries, and one proposal written beside it, are concrete enough to state with the test that would decide them.

4.1 ORDERING ANCHORS BY SYMBOL RARITY

The proposal orders the anchor set by the rarity of the symbol each anchor tests, rarest first, so that the first comparison carries the most rejection power [19]. Comparing a pattern's characters in increasing order of frequency is established in string search, in Sunday's optimal mismatch algorithm [23]. Here it is a change to a function that exists, it costs one sort when the search is planned, and it predicts the direction of a number the bench already prints. On the skewed corpus the first probe refutes about 67 percent of alignments, the figure a collision probability of 0.33 gives.

Hypothesis. Ordering by rarity raises the first probe's refutation rate above that figure. **Test.** The same corpus and patterns, with and without the sort. **Status.** Not run.

4.2 TAIL ERROR UNDER A PERIODIC DISTURBANCE

Introducing a periodic disturbance on one arm lowers the joint error on an undisturbed co-arm by several orders of magnitude. Random distributions with no affine structure work best as the disturbance.

Hypothesis. The error reduction, plotted against the percentage of the tail included, follows a smooth curve, and a straight line would mark a branch that could reach a tail error of 10^{-26}. **Test.** That plot, over disturbances of each kind. **Status.** The plot is not made.

4.3 DOPING AS A KNOWN DISTRIBUTION

A known dopant distribution plays the part a salt plays in cryptography: the hard work of measuring it is done once, and afterwards it is known.

Hypothesis. From measured dopant distributions, new allowed dopant arrangements can be predicted. **Test.** Hold measured arrangements out of the distribution and predict them. **Status.**

Not run. The crystallography research paper reads a published cell edge back from exact site coordinates [18], and no dopant data is read.

4.4 A PROTEIN BACKBONE AS VECTORS AND MAGNITUDES

Ramachandran's allowed regions constrain a protein backbone's dihedral angles [22]. Representing a backbone as a sequence of vectors and magnitudes keeps the order of assembly.

Status. The representation is built. Walking the backbone in assembly order, nitrogen to alpha carbon to carbon, it reads the bond lengths N–C_α, C_α–C and C–N on five structures from 137 to 24,024 bonds. Every length lies within 0.025 Å of the standard values, 1.458, 1.525 and 1.329 Å [16, 20]. The lengths are set by valence, and that makes this a positive control. **A hypothesis, and never tested.** The allowed constructs predict chirality and new allowed proteins.

4.5 USE ON CULTURAL MATERIAL

A culture that was never recorded lives only in its people, passed from one generation to the next. Where they are gone, what was true of it, which statements belong to it, can no longer be determined, though the information need not be gone. A reconstruction presented as true where truth cannot be determined misrepresents the culture it claims to recover.

This work therefore examines only what is present, and asks whether a reconstruction recreated it. The tool is not to be used to regenerate cultural material without a human representative of that culture.

4.6 DATA AND CODE AVAILABILITY

The benches, the instruments and the representations named here are in the orior repository, and their measurements are recorded in the workbook [20].

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